

Unsupervised Learning (clustering)

DATA 202 21FA

Logistics

- Modeling quizzes
- Modeling homework
- Final Projects

Statistical Inference vs Predictive Modeling

Do we need to test for normality etc. for linear regression?

If you want to make **inferences**.

- **Questions about underlying relationships**
 - Is height related to weight?
 - How much faster can you run with caffeine?
 - Does our new website lead to higher sales?
- **Questions about prediction**
 - Guess weight, given height and other details
 - Predict running speed.
 - Predict today's sales.

More Q&A

| When do we use each model?

`parsnip` makes it easy to try many models! But:

- Does it need to be **accurate**? Try many models, probably RF.
- Does it need to be **understandable**? Decision Tree, or sometimes linear regression
 - Very simple shallow trees can be great for decision-makers.
- Will it need to **extrapolate**, e.g., over time? Linear regression.
- etc.

| Don't home prices change year-to-year?

Yeah, we're oversimplifying the Ames examples. Final project?

More Q&A

Why would MAPE be different than MAE?

- Predicting \$5k for a \$1k sale: \$4k absolute error, 400% percent error.
- Predicting \$105k for a \$101k sale: \$4k absolute error, 5% percent error.

Can we change final projects after proposal?

Yes.

Can we use multiple variables in linreg?

Yes! But beware: interpreting coefficients can get tricky.

Unsupervised Learning

- So far we have been doing *supervised* learning, where we have a *target* we're trying to predict.
 - "How much will these homes sell for?"
 - "How long will this person spend watching this video?"
- **Unsupervised** learning works when we don't have an exact target to predict, or we want to explore relationships in the data.
 - "What general types of homes are on the market right now?"
 - "What are some different segments of our customer base?"
 - "Are there distinct types of Covid-19 symptoms?"
- **Clustering** is one very common type of unsupervised learning.

Clustering

Goal: put observations into groups

- Those in the *same* group should be *similar to each other*
- Those in *different* groups should be *different*.

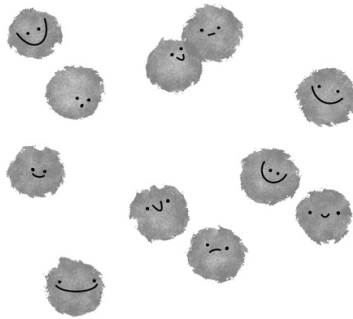
Crucial questions:

- How many groups?
- How do we define "similar" / "different"?

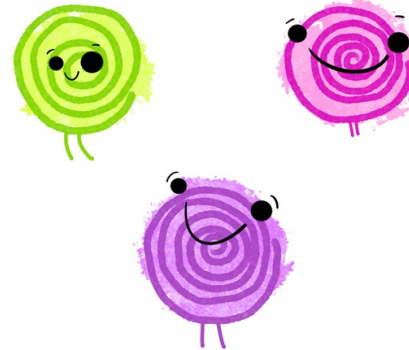
k-means clustering

- assign each observation to one of
- k clusters based on the nearest cluster centroid.

OBSERVATIONS



cluster
CENTROIDS

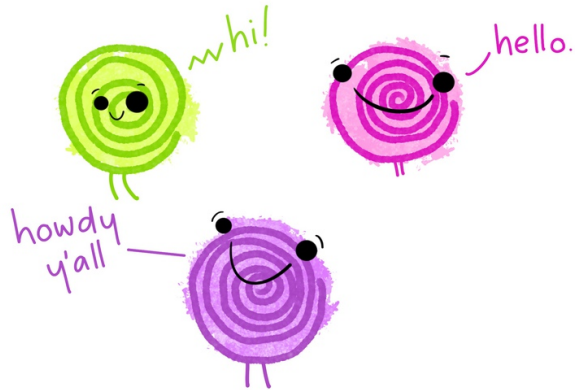


@allison_horst

①

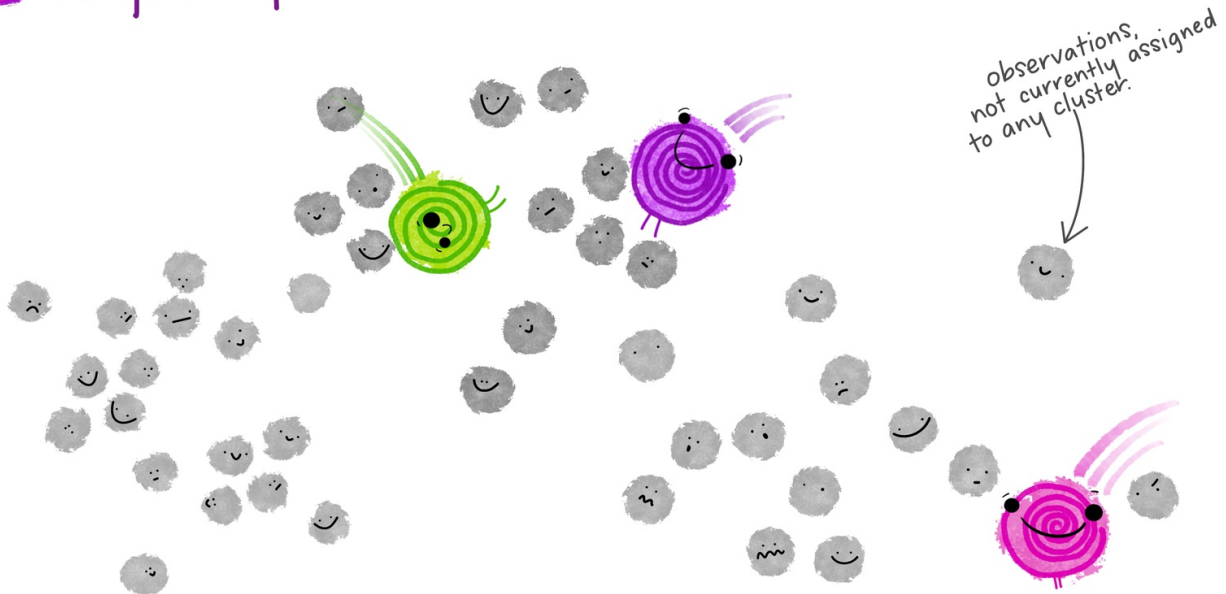
Specify the number of clusters (in this example, $k = 3$).

Then imagine k cluster centroids are created.



②

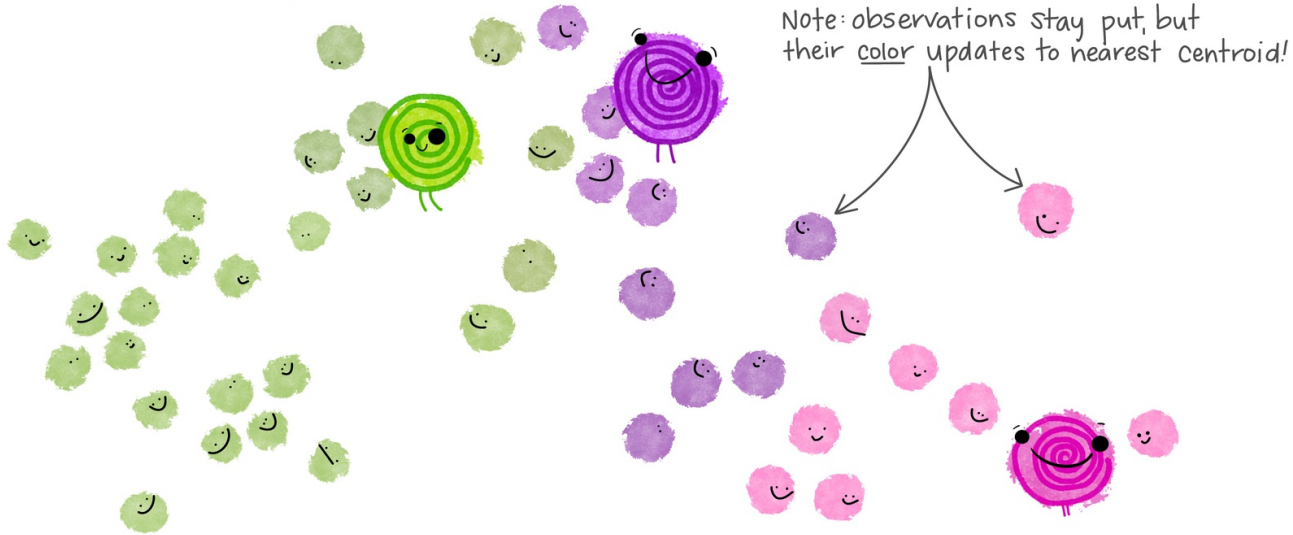
Those k centroids get randomly placed in your space.



③

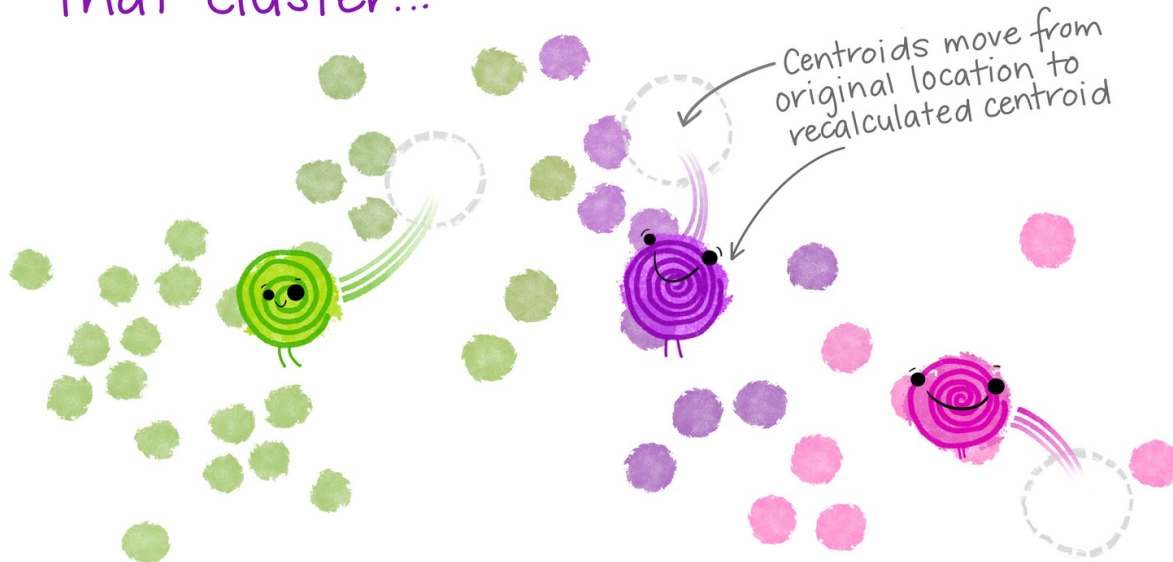
Each observation gets temporarily "assigned" to its closest centroid.

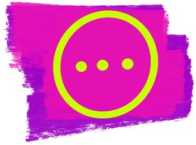
↖ (e.g. by Euclidean distance)



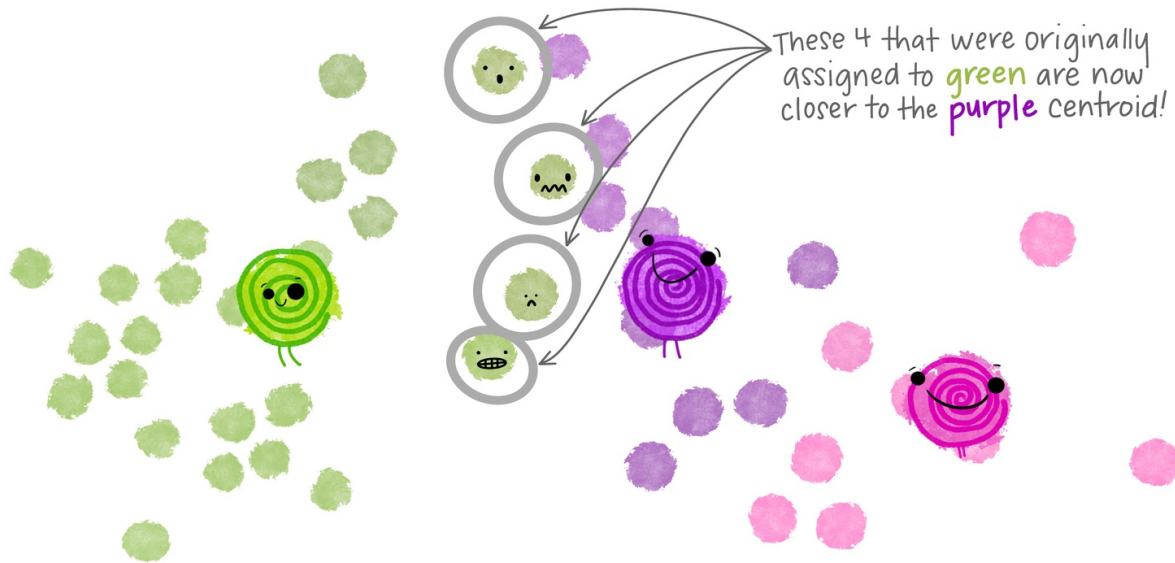
4

Then the centroid of each cluster is calculated based on all observations assigned to that cluster...





UH OH. Now that the cluster centroids have moved, some of the observations are now closer to a different centroid!

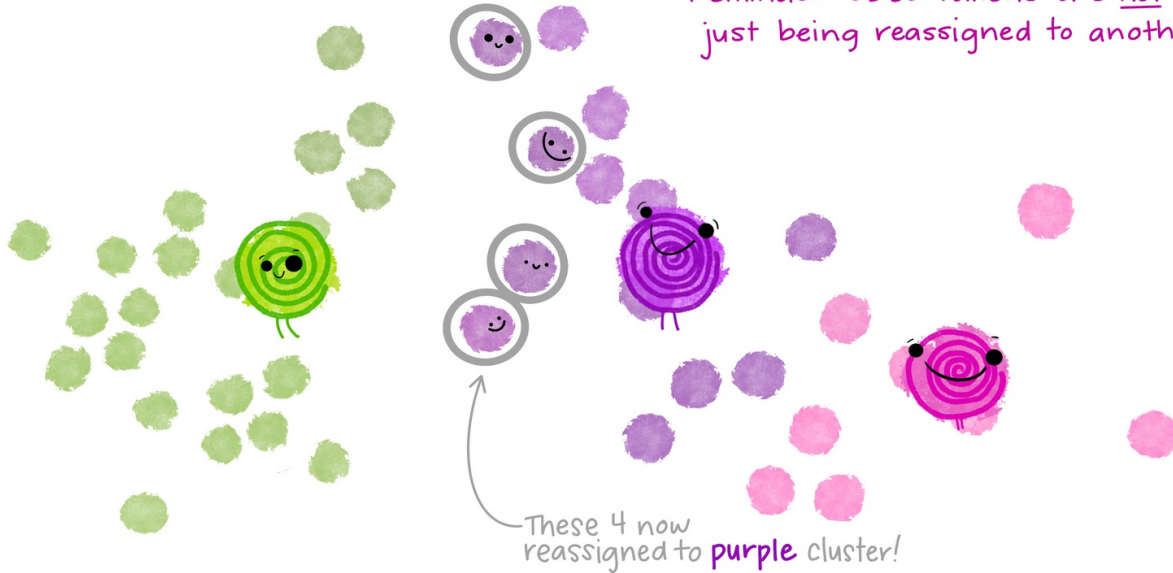




NO PROBLEM!

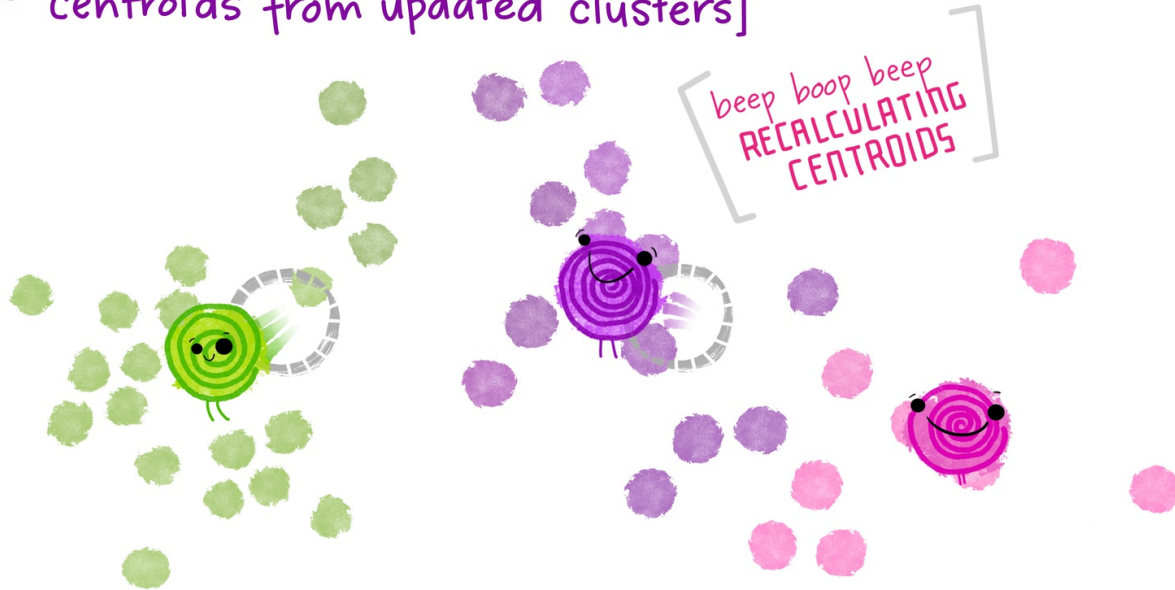
Observations get reassigned* to a different cluster based on the recalculated centroid.

*Reminder: observations are not moving, just being reassigned to another cluster.



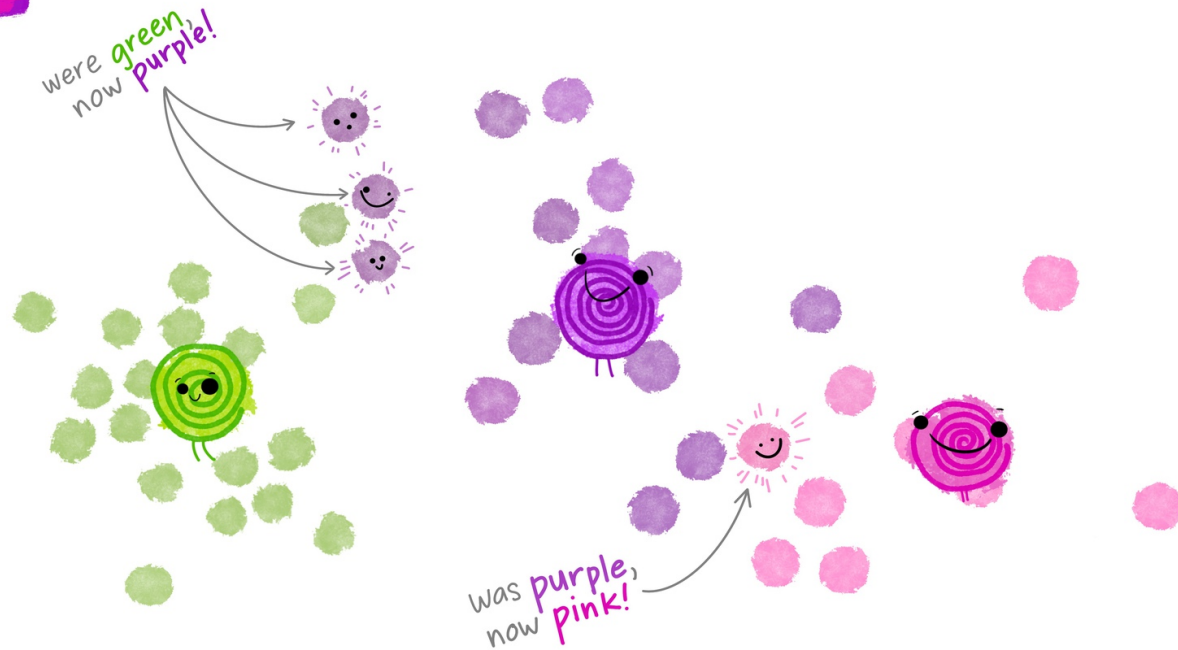
6

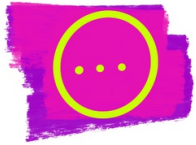
But now that observations have been reassigned, the centroids need to move again [recalculate centroids from updated clusters]



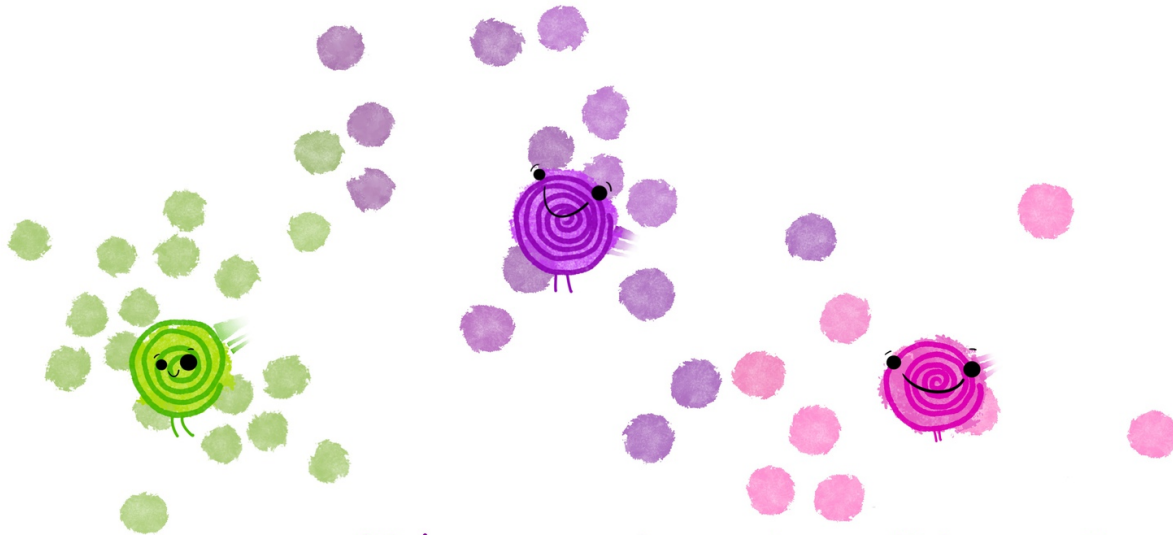
7

Again, now observations are reassigned as needed to the closest centroid.

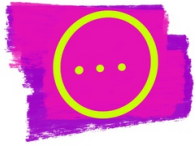




Then the centroid for each cluster
is recalculated...



...which means observations will be reassigned...



That iterative process of

Recalculate cluster centroids

→ Reassign observations to nearest centroid

→ Recalculate cluster centroids

→ Reassign observations to nearest centroid

→ Recalculate cluster centroids

→ Reassign observations to nearest centroid



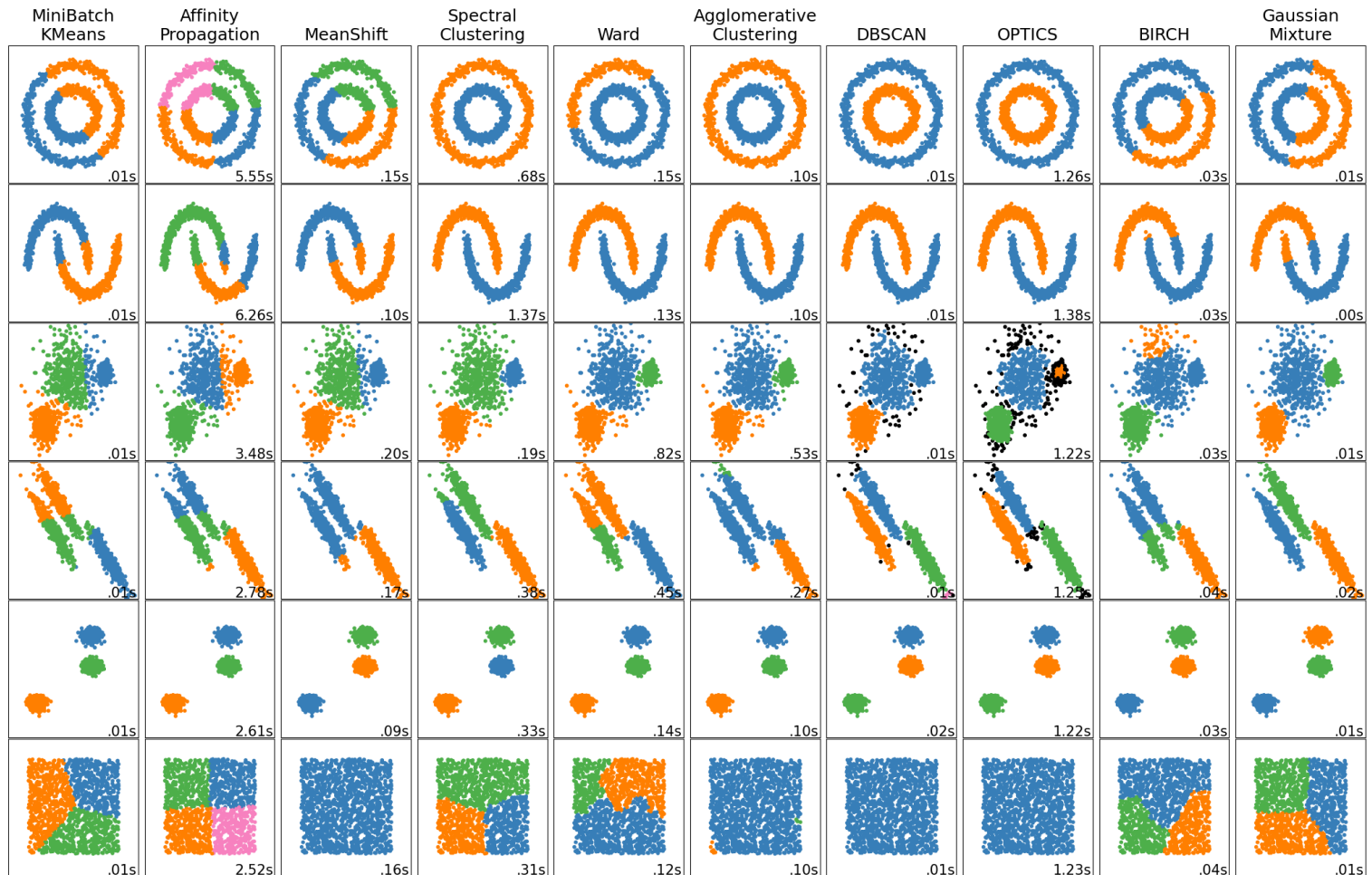
Continues until nothing is moving
or being reassigned anymore!



Which means the iteration is done and each observation is assigned to its final cluster.



Many types of clustering algorithms



```

set.seed(20201120)
data_for_clustering <- ames_all %>%
  select(Latitude, Longitude) %>%
  #select(Year_Built, Gr_Liv_Area) %>%
  mutate(
    #Gr_Liv_Area = rescale(Gr_Liv_Area, to = c(0, 1)),
    #Year_Built = rescale(Year_Built, to = c(0, 1))
  )
clustering_results <- kmeans(
  data_for_clustering, nstart = 4, centers = 3
)

ames_with_clusters <- ames_all %>%
  mutate(cluster = as.factor(clustering_results$cluster))

```

```
glance(clustering_results)
```

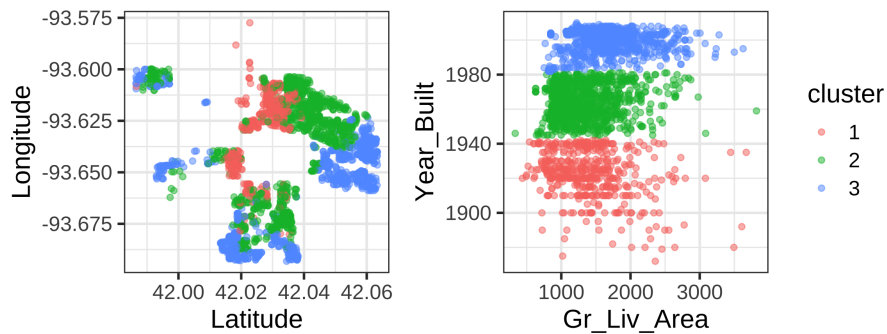
```

# A tibble: 1 × 4
  totss tot.withinss betweenss  iter
  <dbl>      <dbl>      <dbl> <int>
1  2.43      0.687      1.75     3

```

```
tidy(clustering_results)
```

```
latlong_plot <-  
  ggplot(ames_with_clusters, aes(x = Latitude, y = Longitude, color = cluster)) +  
    geom_point(alpha = .5)  
  
year_area_plot <-  
  ggplot(ames_with_clusters, aes(x = Gr_Liv_Area, y = Year_Built, color = cluster)) +  
    #coord_equal() +  
    geom_point(alpha = .5)  
  
library(patchwork)  
latlong_plot + year_area_plot + plot_layout(guides='collect')
```

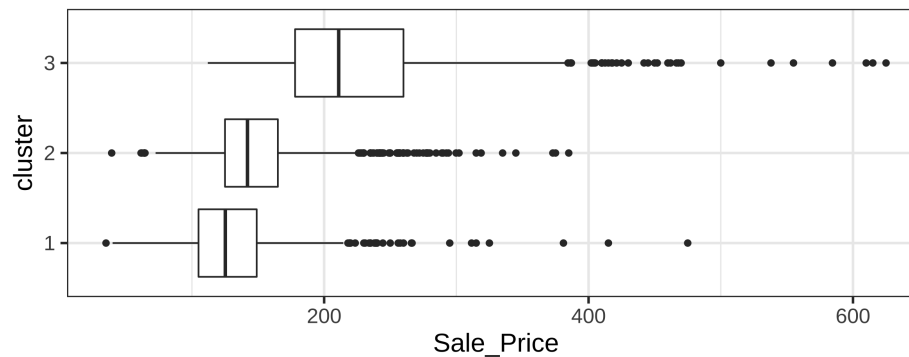


Activities

1. What differences do you notice between the plot on the left and the plot on the right?
2. Try increasing the number of `centers`. What changes about both plots?
3. Use only `Year_Built` for clustering (removing latitude and longitude). What can you say about the age of homes in different parts of town?
4. Try clustering using `select(Latitude, Longitude,`
changes about both plots?
Why are they different?
5. Try scaling `Gr_Liv_Area` to have a maximum of 1 using `Gr_Liv_Area = rescale(Gr_Liv_Area, to = c(0, 1))` etc. What changes about both plots? Why?
6. Try adding scaling for `Latitude` (but not `Longitude`). What changes and why?
7. Now add scaling for `Longitude`. What changes and why?
8. Try changing the maximum

Do the patterns captured by these clusters also happen to relate to sale price?

```
ames_with_clusters %>%  
  ggplot(aes(x = Sale_Price, y = cluster)) + geom_boxplot()
```



Appendix

```
include_graphics("img/sphx_glr_plot_cluster_comparison_001.png")
```

```
#data(ames, package = "modeldata")
ames <- AmesHousing::make_ames()
ames_all <- ames %>%
  filter(Gr_Liv_Area < 4000, Sale_Condition == "Normal") %>%
  mutate(across(where(is.integer), as.double)) %>%
  mutate(Sale_Price = Sale_Price / 1000)
rm(ames)
```